The Skool

How to analyze 

annual report results,

with SQL scripts to determine

future business datadriven decisions

Setup steps

1) Download the annual report from the 

website

2) Save the excel or csv file (the data you want to import) in the same folder you are going to create your SQL scripts

3) Create a SQL script which will load the data

4) Create one SQL scripts per table and start analyzing

Explore the data with these SQL scripts

1A) Amount of applications per educationarea

```
  GNU nano 8.1
SET PAGESIZE 1500
DESC system.results_col3;

SELECT UTBILDNINGSOMRADE AS educationarea,
        SUM(SOKTA_PLATSER_TOTALT) AS applications
FROM results_col3
GROUP BY UTBILDNINGSOMRADE
ORDER BY applications DESC;


EXIT
```

1B) Amount of approved applications per educationarea

```
  GNU nano 8.1
SET PAGESIZE 1500
DESC system.results_col3;


SELECT UTBILDNINGSOMRADE AS educationarea,
       SUM(BEVILJADE_PLATSER_TOTALT) AS approved_applications
FROM results_col3
GROUP BY UTBILDNINGSOMRADE
ORDER BY approved_applications DESC;


EXIT
```

1C) Amount of rejected applications per educationarea

```
 GNU nano 8.1
SET PAGESIZE 1500
DESC system.results_col3;

SELECT UTBILDNINGSOMRADE AS educationarea,
       (SUM(SOKTA_PLATSER_TOTALT)) -
       (SUM(BEVILJADE_PLATSER_UTBILDNINGSO) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS2) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS3) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS4) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS5)) AS rejected_applications
FROM results_col3
GROUP BY UTBILDNINGSOMRADE
ORDER BY rejected_applications DESC;


EXIT
```

Explore the data with these SQL scripts

1D) Amount of approved applications per education within DATA/IT

```
 GNU nano 8.1
SET PAGESIZE 1500
DESC system.results_col3;


SELECT
        UTBILDNINGSNAMN AS title,
       SUM(BEVILJADE_PLATSER_UTBILDNINGSO) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS2) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS3) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS4) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS5) AS approved_applications
FROM results_col3
WHERE UTBILDNINGSOMRADE = 'Data/IT'
       AND UTBILDNINGSNAMN LIKE '%Data%'
       OR UTBILDNINGSNAMN LIKE '%IT%'
GROUP BY UTBILDNINGSOMRADE, UTBILDNINGSNAMN
HAVING SUM(BEVILJADE_PLATSER_UTBILDNINGSO) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS2) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS3) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS4) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS5) > 0
ORDER BY approved_applications DESC, UTBILDNINGSNAMN DESC;


EXIT
```

1E) Amount of rejected applications per education within DATA/IT

```
 GNU nano 8.1
SET PAGESIZE 1500
DESC system.results_col3;


SELECT
        UTBILDNINGSNAMN AS title,
        (SUM(SOKTA_PLATSER_TOTALT)) -
       SUM(BEVILJADE_PLATSER_UTBILDNINGSO) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS2) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS3) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS4) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS5) AS rejected_applications
FROM results_col3
WHERE UTBILDNINGSOMRADE = 'Data/IT'
       AND UTBILDNINGSNAMN LIKE '%Data%'
       OR UTBILDNINGSNAMN LIKE '%IT%'
GROUP BY UTBILDNINGSNAMN
ORDER BY rejected_applications DESC, UTBILDNINGSNAMN DESC;


EXIT
```

Explore the data with these SQL scripts

1F) Amount of approved applications per education provider and education title within DATA/IT

```
 GNU nano 8.1
SET PAGESIZE 1500
DESC system.results_col3;


SELECT UTBILDNINGSANORDNARE_ADMINISTR AS education_provider,
       UTBILDNINGSNAMN AS education,
       SUM(BEVILJADE_PLATSER_UTBILDNINGSO) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS2) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS3) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS4) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS5) AS approved_applications
FROM results_col3
WHERE UTBILDNINGSOMRADE = 'Data/IT'
       AND UTBILDNINGSNAMN LIKE '%Data%'
       OR UTBILDNINGSNAMN LIKE '%IT%'
GROUP BY UTBILDNINGSANORDNARE_ADMINISTR, UTBILDNINGSNAMN
HAVING SUM(BEVILJADE_PLATSER_UTBILDNINGSO) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS2) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS3) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS4) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS5) > 0
ORDER BY approved_applications DESC;


EXIT
```

1G) Top 5 counties, sorted by most approved applications per county

```
 GNU nano 8.1
SET PAGESIZE 1500
DESC system.results_col3;


SELECT LAN AS county,
       SUM(BEVILJADE_PLATSER_UTBILDNINGSO) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS2) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS3) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS4) +
       SUM(BEVILJADE_PLATSER_UTBILDNINGS5) AS approved_applications
FROM results_col3
WHERE LAN <> 'Flera kommuner'
GROUP BY LAN
ORDER BY approved_applications DESC
FETCH FIRST 5 ROWS ONLY;


EXIT
```

Explore the data with these SQL scripts

1H) Percentages of remote and on-campus programs within each educationarea

```
  GNU nano 8.1
SET PAGESIZE 1500
DESC system.results_col3;


SELECT
    rl.utbildningsomrade AS educationarea,
    SUM(CASE
        WHEN rl.STUDIEFORM = 'Bunden'
        THEN rl.Sokta_platser_totalt
        ELSE 0
    END) AS campus_applications,
    SUM(CASE
        WHEN rl.STUDIEFORM = 'Distans'
        THEN rl.Sokta_platser_totalt
        ELSE 0
    END) AS remote_applications,
    ROUND(
        100 * SUM(CASE WHEN rl.STUDIEFORM = 'Bunden'
            THEN rl.Sokta_platser_totalt ELSE 0 END)
        / NULLIF(SUM(rl.Sokta_platser_totalt), 0), 2
    ) || '%' AS campus_applications_percentages,
    ROUND(
        100 * SUM(CASE WHEN rl.STUDIEFORM = 'Distans'
            THEN rl.Sokta_platser_totalt ELSE 0 END)
        / NULLIF(SUM(rl.Sokta_platser_totalt), 0), 2
    ) || '%' AS remote_applications_percentages,
    SUM(rl.Sokta_platser_totalt) AS total_applications
FROM results_col3 rl
GROUP BY
    rl.utbildningsomrade
ORDER BY
    rl.utbildningsomrade DESC;


EXIT
```

1I) Amount of educations including the word ”AI”

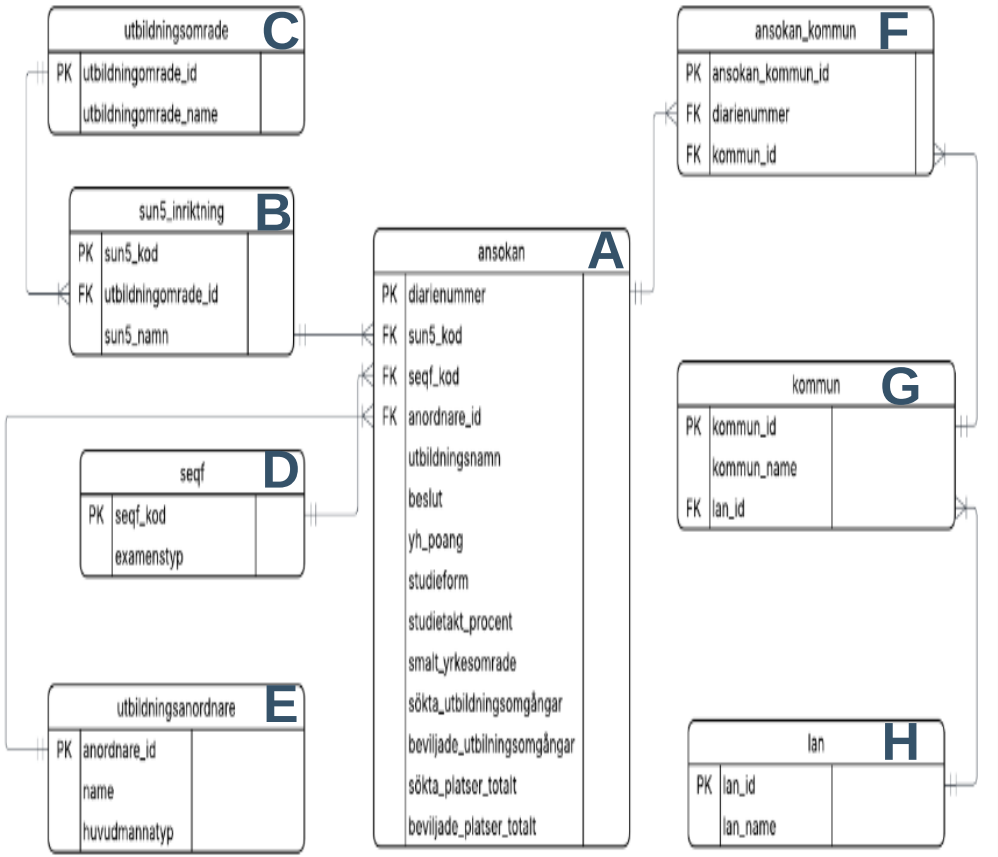
```
  GNU nano 8.1
SET PAGESIZE 1500
DESC system.results_col3;


SELECT UTBILDNINGSNAMN AS education_title
        FROM results_col3
WHERE (UTBILDNINGSNAMN LIKE '%AI%'
        OR UTBILDNINGSNAMN LIKE '%ai%');


SELECT
        COUNT(*) AS amount_of_educations_including_AI
        FROM results_col3
WHERE (UTBILDNINGSNAMN LIKE '%AI%'
        OR UTBILDNINGSNAMN LIKE '%ai%');


EXIT
```

CTAS and JOINS




```
 GNU nano 8.1                                                    lJ.sql
SELECT
        c.UTBILDNINGSOMRADE AS educationarea,
        b.SUN5_INRIKTNING_NAMN AS specialization,
        a.UTBILDNINGSNAMN AS education,
        a.STUDIEFORM AS studyform,
        a.SOKTA_OMGANGAR AS applied_rounds,
        a.BEVILJADE_OMGANGAR AS approved_rounds,
        a.SOKTA_PLATSER_TOTALT AS applied_spots,
        a.BEVILJADE_PLATSER_TOTALT AS approved_spots,
        e.ANORDNARE AS education_provider,
        g.KOMMUN AS municipality,
        h.LAN AS county,
        f.DIARIENUMMER AS reference_id,
        d.SEQF,
        a.SMALT_YRKE AS narrow_occupation
FROM ANSOKAN a
        JOIN SUN5_INRIKTNING b ON b.SUN5_ID = a.SUN5_ID
        JOIN UTBILDNINGSOMRADE c ON c.UTBILDNINGSOMRADE_ID = b.UTBILDNINGSOMRADE_ID
        JOIN SEQF d ON d.SEQF_ID = a.SEQF_ID
        JOIN ANORDNARE e ON e.ANORDNARE_ID = a.ANORDNARE_ID
        JOIN ANSOKAN_KOMMUN f ON f.ANSOKAN_KOMMUN_ID = a.ANSOKAN_KOMMUN_ID
        JOIN KOMMUN g ON g.KOMMUN_ID = f.KOMMUN_ID
        JOIN LAN h ON h.LAN_ID = g.LAN_ID
WHERE c.UTBILDNINGSOMRADE = 'Data/IT'
        AND a.UTBILDNINGSNAMN LIKE '%Data%'
        OR a.UTBILDNINGSNAMN LIKE '%IT%'
GROUP BY c.UTBILDNINGSOMRADE,
        b.SUN5_INRIKTNING_NAMN,
        a.UTBILDNINGSNAMN,
        a.STUDIEFORM,
        a.SOKTA_OMGANGAR,
        a.BEVILJADE_OMGANGAR,
        a.SOKTA_PLATSER_TOTALT,
        a.BEVILJADE_PLATSER_TOTALT,
        e.ANORDNARE,
        g.KOMMUN,
        h.LAN,
        f.DIARIENUMMER,
        d.SEQF,
        a.SMALT_YRKE
HAVING (SUM(a.BEVILJADE_PLATSER_TOTALT)) > 0
ORDER BY a.BEVILJADE_OMGANGAR DESC, a.BEVILJADE_PLATSER_TOTALT DESC;

EXIT
```

Additional useful SQL scripts

3A) Most approved educationsareas, sorted by the highest applications rate

```
 GNU nano 8.1
SET PAGESIZE 1500
DESC system.results_col3;

SELECT
        UTBILDNINGSOMRADE AS educationarea,
        COUNT(*) AS total_applications,
        SUM(CASE WHEN Beslut = 'Beviljad'
                THEN 1 ELSE 0 END) AS approved_applications,
        SUM(CASE WHEN Beslut = 'Avslag'
                THEN 1 ELSE 0 END) AS rejected_applications,
        ROUND(
        100 * SUM(CASE WHEN Beslut = 'Beviljad'
                THEN 1 ELSE 0 END) / COUNT(*), 2
        ) || '%' AS approvals_in_percentages
FROM results_col3
GROUP BY UTBILDNINGSOMRADE
ORDER BY approvals_in_percentages DESC;

EXIT
```

3B) Most approved SUN5 specializations within transportationservices, sorted by the highest applications rate

```
 GNU nano 8.1
SET PAGESIZE 1500
DESC system.results_col3;

SELECT
        SUN5_INRIKTNING AS SUN5_code,
        SUN5_INRIKTNING_NAMN AS SUN5_specialization,
        COUNT(*) AS applications,
        SUM(CASE WHEN Beslut = 'Beviljad'
                THEN 1 ELSE 0 END) AS approved,
        SUM(CASE WHEN Beslut = 'Avslag'
                THEN 1 ELSE 0 END) AS rejected,
        ROUND(
        100 * SUM(CASE WHEN Beslut = 'Beviljad'
                THEN 1 ELSE 0 END) / COUNT(*), 2
        ) || '%' AS approvals_in_percentages
FROM results_col3
WHERE (UTBILDNINGSOMRADE LIKE '%Transporttj_nster%')
GROUP BY
        SUN5_INRIKTNING,
        SUN5_INRIKTNING_NAMN
ORDER BY (ROUND(
        100 * SUM(CASE WHEN Beslut = 'Beviljad'
                THEN 1 ELSE 0 END) / COUNT(*), 2
        )) DESC;

EXIT
```

Additional useful SQL scripts

```
 GNU nano 8.1
SET PAGESIZE 1500
DESC system.results_col3;

SELECT
        DIARIENUMMER AS reference_id,
        UTBILDNINGSNAMN AS education,
        SMALT_YRKESOMRADE AS narrow_occupation,
        STUDIEFORM AS studyform,
        UTBILDNINGSANORDNARE_ADMINISTR AS education_provider,
        HUVUDMANNATYP AS type,
        KOMMUN AS municipality,
        BEVILJADE_UTBILDNINGSOMGANGAR AS approved_rounds,
        COUNT(*) AS applications,
        SUM(CASE WHEN Beslut = 'Beviljad'
                THEN 1 ELSE 0 END) AS approved,
        SUM(CASE WHEN Beslut = 'Avslag'
                THEN 1 ELSE 0 END) AS rejected,
        ROUND(
        100 * SUM(CASE WHEN Beslut = 'Beviljad'
                THEN 1 ELSE 0 END) / COUNT(*), 2
        ) || '%' AS approvals_percentages
FROM results_col3
WHERE (SUN5_INRIKTNING = '840aa'
        OR SUN5_INRIKTNING LIKE '%840cy%')
GROUP BY
        DIARIENUMMER,
        UTBILDNINGSNAMN,
        SMALT_YRKESOMRADE,
        STUDIEFORM,
        UTBILDNINGSANORDNARE_ADMINISTR,
        HUVUDMANNATYP,
        KOMMUN,
        BEVILJADE_UTBILDNINGSOMGANGAR
ORDER BY approved_rounds DESC, approvals_percentages DESC;

EXIT
```

<-- 3C) Most approved educations within transportationservices, sorted by most approved rounds and the highest applications rate

3D) -->
Most approved SUN5 specializations within DATA/IT, sorted by the highest applications rate

```
 GNU nano 8.1
SET PAGESIZE 1500
DESC system.results_col3;

SELECT
        SUN5_INRIKTNING AS SUN5_code,
        SUN5_INRIKTNING_NAMN AS SUN5_specialization,
        COUNT(*) AS applications,
        SUM(CASE WHEN Beslut = 'Beviljad'
                THEN 1 ELSE 0 END) AS approved,
        SUM(CASE WHEN Beslut = 'Avslag'
                THEN 1 ELSE 0 END) AS rejected,
        ROUND(
        100 * SUM(CASE WHEN Beslut = 'Beviljad'
                THEN 1 ELSE 0 END) / COUNT(*), 2
        ) || '%' AS approvals_in_percentages
FROM results_col3
WHERE utbildningsomrade = 'Data/IT'
        AND Beslut IN ('Beviljad', 'Avslag')
GROUP BY
        SUN5_INRIKTNING,
        SUN5_INRIKTNING_NAMN
ORDER BY (ROUND(
        100 * SUM(CASE WHEN Beslut = 'Beviljad'
                THEN 1 ELSE 0 END) / COUNT(*), 2
        )) DESC;

EXIT
```

Additional SQL scripts

3E) -->

Most approved educations within DATA/IT, sorted by most approved rounds and the highest applications rate

```
  GNU nano 8.1
SET PAGESIZE 1500
DESC system.results_col3;

SELECT
        DIARIENUMMER AS reference_id,
        UTBILDNINGSNAMN AS education,
        SMALT_YRKESOMRADE AS narrow_occupation,
        STUDIEFORM AS studyform,
        UTBILDNINGSANORDNARE_ADMINISTR AS education_provider,
        HUVUDMANNATYP AS type,
        KOMMUN AS municipality,
        BEVILJADE_UTBILDNINGSOMGANGAR AS approved_rounds,
        COUNT(*) AS applications,
        SUM(CASE WHEN Beslut = 'Beviljad'
                THEN 1 ELSE 0 END) AS approved,
        SUM(CASE WHEN Beslut = 'Avslag'
                THEN 1 ELSE 0 END) AS rejected,
        ROUND(
        100 * SUM(CASE WHEN Beslut = 'Beviljad'
                THEN 1 ELSE 0 END) / COUNT(*), 2
        ) || '%' AS approvals_percentages
FROM results_col3
WHERE (SUN5_INRIKTNING = '481ae'
        OR SUN5_INRIKTNING LIKE '%481ab%')
GROUP BY
        DIARIENUMMER,
        UTBILDNINGSNAMN,
        SMALT_YRKESOMRADE,
        STUDIEFORM,
        UTBILDNINGSANORDNARE_ADMINISTR,
        HUVUDMANNATYP,
        KOMMUN,
        BEVILJADE_UTBILDNINGSOMGANGAR
ORDER BY approved_rounds DESC, approvals_percentages DESC;

EXIT
```

Recommendations

DATA / IT

- **Apply for Game Programmer / Spelutvecklare, 3 rounds, on-campus, Stockholm**
- **Apply for Systemutvecklare DevSecOps, 2 rounds, on-campus, multiple muncipalities**
- **Apply for Database Administrator – DBA, 2 rounds, remote, Stockholm**

Recommendations

DATA / IT

- **Apply for Game Programmer / Spelutvecklare, 3 rounds, on-campus, Stockholm**
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- **Apply for Database Administrator – DBA, 2 rounds, remote, Stockholm**

Do a competitor analyze to distingish your educations from the ones that already exists!

Difference between spelutvecklare, GP & GP Specialization? What can you offer?

Connection to location? Can you offer similar education in different locations or remote?